

When Does Answer Diversity Mean Disagreement?

Possible Worlds and the Scope of Semantic Entropy

Auxiliary report to the paper *Who Said It? Speaker-Conditioned Verification and Lexicographic Tie-Breaking for Low-Cost Hallucination Detection in Legislative Hearings*

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Abstract

Abstract. Sampling-based uncertainty methods such as semantic entropy group a model’s sampled answers into classes of equivalent meaning and read many classes as a sign that the model does not know the answer. This report, which accompanies our paper on hallucination detection in PublicHearingBR, examines a distinction these methods do not make: two answers can differ because they contradict each other or because both can be true. Using the contradiction probabilities that the natural language inference model already computes, we represent an answer set as an event structure whose maximal consistent subsets, its possible worlds, can be enumerated exactly. We show that semantic entropy and every quantity derived from the entailment order are blind to this distinction, and we measure it in three settings. On TriviaQA, the model’s answer is correct 81% of the time when its samples share one meaning, 44% when they differ but fit in a single world ($n = 18$) and 22% when they span several worlds ($n = 125$), although the last two rates do not differ significantly. The share of multi-class answer sets that form a single world, which we call the conflation rate, is 12.6% in short-answer question answering and 78.5% in closed-book statements about hearings. Asking the same TriviaQA questions for full-sentence answers raises the conflation rate to 29.0% and lowers the Area Under the Receiver Operating Characteristic Curve (AUROC) of semantic entropy from 0.816 to 0.721 (paired difference -0.095 , 95% CI $[-0.175, -0.016]$). Entropy over worlds does not outperform semantic entropy where the latter works, and consistency features do not detect hallucinations in PublicHearingBR; we report both negative results in full.

Keywords: Semantic entropy; Natural language inference; Contradiction; Possible worlds; Event structures; Uncertainty quantification; Hallucination detection

1 Introduction

This report accompanies the paper *Who Said It? Speaker-Conditioned Verification and Lexicographic Tie-Breaking for Low-Cost Hallucination Detection in Legislative Hearings*, submitted to the first data challenge of Instituto Kunumi. The paper checks opinions attributed to participants of legislative hearings against the attributed speaker’s own turns and combines inexpensive detectors with a committee of LLM judges. A second line of our project studied a different question, which the paper leaves out for reasons of space and scope: when a language model’s sampled answers differ, how much of the difference is disagreement? This report presents that study, including its negative results. It can be read independently of the paper; Section 7 contains the one result the paper cites.

A common way to estimate whether a language model knows an answer is to sample it several times and measure how much the samples differ [9, 11]. *Semantic entropy* [5, 9] groups the samples into classes of answers that entail each other, according to a natural language inference (NLI) model, and computes the entropy of the class frequencies. One class means that the samples agree; many classes are read as a sign that the model does not know. On short-answer question answering this works well: on TriviaQA [7], semantic entropy reaches an Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.811 in our setup (Section 6.2).

Two answers can fall into different classes for three reasons. They can contradict each other (“the committee approved the bill” and “the committee rejected the bill”); one can refine the other (“the committee approved the bill by a wide margin”); or they can state different facts that are both true (“the session lasted four hours”). Only the first is a disagreement about what happened. Semantic entropy counts all three alike, although the NLI model it already runs returns a contradiction probability for every pair of answers. Methods that do use contradiction treat it as a score, such as an average or a density (Section 2.3), rather than as a constraint on which answers can be true together.

We use that constraint directly. Section 3 represents an answer set as an event structure [15]: answers ordered by entailment and related by conflict. Its maximal consistent subsets, which we call *worlds*, are the complete accounts the samples support, and they can be enumerated exactly. Section 4 shows that semantic entropy and every quantity computed from the entailment order are blind to conflict, and Sections 5–7 measure what this blindness costs. Our contributions are:

- a formal account of which answers can hold together (Proposition 3), with an entropy over worlds that needs no model calls beyond the NLI passes semantic entropy already makes;
- evidence that conflict carries information the entailment order does not: on TriviaQA, 97.9% of conflicts join classes the order does not compare, and the accuracy of the model’s answer falls from 81.1% (one class) to 44.4% (several classes in one world, $n = 18$) and 22.4% (several worlds), although the last step is not significant;
- the *conflation rate*, a statistic computed without labels, which is 12.6% on TriviaQA and 49.3–78.5% for closed-book statements about hearings, together with an intervention on the same TriviaQA questions in which a higher conflation rate comes with a significant drop in the AUROC of semantic entropy;
- negative results: the entropy over worlds does not outperform semantic entropy where the latter works; in PublicHearingBR, contradictions among the retrieved evidence carry no detectable information about hallucination beyond the amount of evidence; and none of the self-consistency quantities we tested, ours included, detects answers built on misleading evidence.

All predictions were written to the repository before the corresponding measurement (Section 8). Results outside the registered confirmatory comparisons are labelled exploratory.

2 Background

2.1 Natural language inference

An NLI model reads an ordered pair of sentences, a premise and a hypothesis, and returns three probabilities that sum to one (Table 1). Entailment is directed: “it was in 1988” entails “it was in the 1980s”, but not conversely. Contradiction is symmetric in meaning, although the model’s estimates for the two orders of a pair can differ. Throughout this report we use the multilingual checkpoint `mDeBERTa-v3-base-xnli-multilingual-nli-2mil7` [10], with a threshold of 0.5 for both entailment and contradiction, fixed before any measurement.

Table 1. The three outputs of an NLI model.

Label	Meaning	Example (premise → hypothesis)
entailment	the premise makes the hypothesis true	“it was in 1988” → “it was in the 1980s”
contradiction	the two cannot both be true	“it was in 1988” → “it was in 1995”
neutral	neither of the above	“it was in 1988” → “it rained that day”

2.2 Semantic entropy and the entailment order

Consider N answers sampled for the same prompt. Write $a \sqsubseteq b$ when a entails b with probability at least 0.5, closed under transitivity (the NLI model is not transitive by itself). Answers that entail each other form a *semantic class*, and the mass $p(u)$ of class u is the fraction of samples in it. **Semantic entropy** is $SE = -\sum_u p(u) \log p(u)$, the discrete estimator of Farquhar et al. [5]. It is 0 when all samples share one meaning and $\log N$ when no two do.

Entailments in one direction only survive the grouping and order the classes: $u \sqsubseteq v$ means that u entails v , so v is the more general class. A *chain* is a set of classes stacked one above the other, such as “in 1988” $\sqsubseteq$ “in the 1980s” $\sqsubseteq$ “in the 20th century”: one belief at three levels of detail. An *antichain* is a set of classes no two of which are comparable. Semantic entropy ignores the order: the chain above and three unrelated answers both give $\log 3$.

As an order-aware comparator we also report the **core entropy** SE_{core} , which we developed earlier in this project. A class is *redundant* (a *beat point*, 14) if the classes strictly below it have a greatest element or the classes strictly above it have a least element. Removing redundant classes one at a time, and moving the mass of each removed class to the class that witnessed its removal, ends in the *core* of the order, which is unique up to isomorphism [14]. SE_{core} is the entropy of the core’s masses, averaged over eight removal orders because the final masses, unlike the core, can depend on the order. On a chain the core is a single class and $SE_{\text{core}} = 0$; on an antichain nothing is removed and $SE_{\text{core}} = SE$.

2.3 How contradiction is used elsewhere

SelfCheckGPT [11] can score a sentence by the average probability that sampled passages contradict it. Kernel language entropy [12] replaces hard classes by a similarity kernel built from NLI outputs; directional entailment graphs [2] keep the direction of entailment; and logical graphs [3] combine the entailment structure with a scalar density of contradictions. In these methods contradiction enters as a number (an average, a weight or a density). To our knowledge, none of them uses it to determine which answers can be true together, which is the object of this report.

3 Worlds

This section defines which subsets of an answer set can be held together, first informally and then formally. Throughout, P is the set of semantic classes of one answer set, ordered by $\sqsubseteq$, with masses p .

3.1 Conflict and its inheritance

Two classes u and v are in *raw conflict*, written $u \#_0 v$, if some answer of u and some answer of v contradict each other with probability at least 0.5, in either order. Contradictions between answers of the same class are ignored. Raw conflict needs one correction. If “the committee approved the bill” contradicts “the committee rejected the bill”, then “the committee approved the bill by a wide margin” contradicts it as well, since anyone who asserts the refinement also asserts what it entails. The NLI model may miss such inherited contradictions, so we add them.

Definition 1 (Inherited conflict). Classes u and w are in *conflict*, written $u \# w$, if they have generalisations in raw conflict: $u \sqsubseteq v$, $w \sqsubseteq v'$ and $v \#_0 v'$ for some classes v, v' , where every class counts as a generalisation of itself. If P and C_0 denote the Boolean matrices of $\sqsubseteq$ and $\#_0$, the conflict matrix is $C^* = \text{bool}(P C_0 P^\top)$.

Conflict is symmetric and is inherited by refinements: if $v \# w$ and $u \sqsubseteq v$, then $u \# w$. A class can be in conflict with itself, when two of its generalisations are in raw conflict. No one can hold such a class consistently, so we exclude it from every world, drop its mass, and report the share of such classes as a diagnostic of the NLI model (the *self-conflict rate*). The share of class pairs that inheritance adds to raw conflict is the *closure cost*.

3.2 Worlds

Informally, a world is a largest set of answers that can all be true at once. A consistent belief must also respect the order: whoever holds a class holds everything it entails.

Definition 2 (Belief state, world). A *belief state* is a set S of classes that contains no self-conflicting class and no pair in conflict, and is closed under generalisation: if $u \in S$ and $u \sqsubseteq v$, then $v \in S$. A *world* is a belief state that is not contained in any larger belief state.

In the terminology of concurrency theory, $(P, \sqsubseteq, \#)$ is an event structure [15] in which generalisation plays the role of causal dependency, and belief states are its configurations. Inheritance of conflict is the axiom event structures require. It is also what makes worlds easy to compute, because it allows the closure condition to be dropped.

Proposition 3 (Worlds are maximal independent sets). *Call a set of classes independent if it contains no self-conflicting class and no pair in conflict. The worlds are exactly the maximal independent sets. In particular, every maximal independent set is closed under generalisation.*

Proof. Two facts follow from Definition 1. (a) If u is not self-conflicting and $u \sqsubseteq v$, then v is not self-conflicting, because every generalisation of v is a generalisation of u . (b) If $u \sqsubseteq v$ and $v \# w$, then $u \# w$.

Let W be a maximal independent set, $u \in W$ and $u \sqsubseteq v$. By (a), v is not self-conflicting. If $v \notin W$, maximality gives some $w \in W$ with $v \# w$, and (b) gives $u \# w$, contradicting independence. Hence W is closed under generalisation and is a belief state. Every belief state is independent, so no belief state strictly contains W , and W is a world.

Conversely, let S be a world and suppose that $S \cup \{x\}$ is independent for some class $x \notin S$. Let $S' = S \cup \{v : x \sqsubseteq v\}$. If a generalisation v of x were in conflict with some $s \in S$, (b) would give $x \# s$; if two generalisations of x were in conflict, x would be self-conflicting; and by (a) no generalisation of x is self-conflicting. So S' is independent, and it is closed under generalisation because S is. Then S' is a belief state strictly containing S , a contradiction. Hence S is a maximal independent set. $\square$

The proposition turns world enumeration into a standard graph problem: the worlds are the maximal cliques of the *consistency graph*, whose vertices are the classes that are not self-conflicting and whose edges join classes not in conflict. We enumerate them with the Bron–Kerbosch algorithm with pivoting [1]; answer sets here have at most ten classes, and enumeration takes microseconds. We also checked the implementation numerically: on 400 random graphs with 3 to 10 vertices, enumeration agrees with brute force, and on 400 random orders with random conflicts all 918 worlds found are closed under generalisation.

3.3 Entropy over worlds, conflict density and conflation

To compare worlds with semantic entropy we distribute the class masses over the worlds. Each class splits its mass equally among the worlds that contain it: if $m(u)$ is the number of worlds containing u ,

$$q(W) \propto \sum_{u \in W} \frac{p(u)}{m(u)}, \quad \text{WE} = - \sum_W q(W) \log q(W), \quad (1)$$

with q normalised to sum to one (the mass of self-conflicting classes is dropped). We call WE the **world entropy**. Three properties guide its reading. If no two classes are in conflict, all classes form a single world and $\text{WE} = 0$, whatever the value of SE. If the classes are pairwise incomparable and every pair is in conflict, the worlds are the classes themselves and $\text{WE} = \text{SE}$ (a conflict between comparable classes makes the more specific one self-conflicting instead). And WE is not a discounted SE: with six unrelated classes of equal mass and three disjoint conflicts ($a \# b$, $c \# d$, $e \# f$), there are eight equally weighted worlds and $\text{WE} = \log 8 > \text{SE} = \log 6$.

We use two further summaries. The **conflict density** of an answer set is the share of its class pairs in conflict. It is a scalar in the spirit of Dong et al. [3] and serves as a control: if worlds carried no more information than the amount of conflict, they would not justify their definition. The **conflation rate** of a collection of answer sets is the share of sets with more than one class that nonetheless form a single world. In such a set no two classes that can be held consistently contradict each other: any contradiction it contains involves a self-conflicting class. Its diversity is elaboration or complementary detail rather than disagreement.

3.4 A worked example

Asked what a hearing concluded, a model returns four answers (Figure 1a): a_1 “the committee approved the bill”, a_2 “the committee approved the bill by a wide margin”, a_3 “the committee rejected the bill” and a_4 “the session lasted four hours”. Only $a_2 \sqsubseteq a_1$ holds, so there are four classes of mass $1/4$ and $\text{SE} = \log 4 = 1.386$, the maximum for four samples. The NLI model reports $a_1 \# a_3$, and inheritance adds $a_2 \# a_3$. The worlds are $\{a_1, a_2, a_4\}$ and $\{a_3, a_4\}$; a_4 splits its mass between them, so their masses

are $5/8$ and $3/8$ and $WE = 0.662$. Of the three distinctions semantic entropy counts, only one is a disagreement: a_2 elaborates a_1 , and a_4 is about something else. Without a_3 , semantic entropy would still report $\log 3 = 1.099$ for a set containing no disagreement at all, while $WE = 0$.

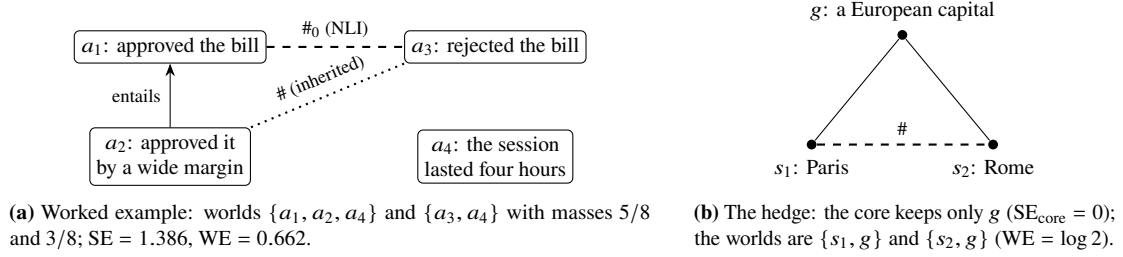


Figure 1. Conflict, inheritance and worlds. Solid lines are entailments (the upper class is more general), dashed lines are contradictions reported by the NLI model, and the dotted line is a contradiction added by inheritance.

4 What an entropy over classes cannot see

Observation 4 (Conflict blindness). *Semantic entropy, the core entropy, the number of classes, the size of the core, the height of the order, and any other quantity computed only from the classes, their masses and the entailment order do not change when the conflict relation changes. In particular, for every number of classes $n \geq 2$ there are two answer sets with n classes on which all these quantities coincide, one with a single world and one with n worlds.*

Proof. None of these quantities takes the conflict relation as input. For the second statement, take n pairwise incomparable classes of equal mass, once with no conflict and once with every pair in conflict. Classes, masses and order coincide; the first set has one world, containing every class, and the second has n worlds, one for each class. $\square$

The observation is immediate; its role is to state precisely what the entailment order leaves out. Its two answer sets are not artificial. Six answers to “what did the hearing conclude?” can be six compatible aspects of one conclusion or six incompatible conclusions, and Section 6.3 shows that the first case covers most multi-class answer sets when Qwen is asked, without the transcript, about a participant’s position in a hearing, and about half of them with Llama. Table 2 adds three configurations that separate the quantities further.

Table 2. Constructed answer sets of six samples in which the quantities disagree (script event_structure). SE: semantic entropy; SE_{core} : core entropy; worlds and WE as in Section 3.

Answer set	SE	SE_{core}	Worlds	WE
Chain: one belief at six levels of detail, no conflict	1.792	0.000	1	0.000
Antichain of six classes, no conflict	1.792	1.792	1	0.000
Antichain of six classes, every pair in conflict	1.792	1.792	6	1.792
Two specific classes in conflict, each below the same two general ones, plus two unrelated classes	1.792	1.792	2	0.693
Hedge: $s_1 \# s_2$ below one general class g	1.099	0.000	2	0.693

The hedge. A model that does not know an answer often hedges: across samples it names two incompatible specifics and one safe generalisation (“Paris”, “Rome”, “a European capital”). The generalisation is entailed by both specifics, which contradict each other (Figure 1b). Each specific has a single class above it, so the core removes both and reports $SE_{core} = 0$: no uncertainty for an answer set whose content is uncertainty. The worlds keep the disagreement: $\{s_1, g\}$ and $\{s_2, g\}$, with $WE = \log 2$. This is the smallest configuration in which removing redundancy from the order destroys the signal. Conflicts between two classes with a common generalisation occur in 6.4% of the TriviaQA answer sets, which is consistent with the small, non-significant loss of the core entropy against semantic entropy there (Section 6.2).

The topology of conflict. The independent sets form a simplicial complex, the independence complex $\text{Ind}(C^*)$ of the conflict graph [8], whose maximal faces are the worlds. By Dowker’s theorem [4], the nerve of the cover of the classes by worlds has the same homotopy type. Its number of connected components b_0 counts blocks of classes that conflict with every class outside their block. Its first Betti number b_1 can be positive only if the consistency graph contains an induced cycle of length at least four. The smallest example is two independent binary disagreements ($a \# b$ and $c \# d$), which give four worlds and $b_1 = 1$ without any cycle of conflicts. (An earlier draft required a cycle of five conflicting classes, which is the smallest case only among conflict graphs that are cycles.) Our implementation reproduces the known homotopy types of $\text{Ind}(C_n)$ for $n = 4, \dots, 7$ [8]. We measure $b_1 > 0$ in at most 5.8% of the answer sets of any setting and do not use it as a signal.

5 Data and pipeline

Table 3 lists the answer sets. All samples were generated by open-weight models served with Ollama, at temperature 1.0.

- **TriviaQA, entity answers.** 233 questions from the validation split without context [7], answered with only the entity. The label is whether a low-temperature answer ($T = 0.1$) contains a normalised gold alias; no neural model is involved.
- **TriviaQA, sentence answers.** The same questions, asking for one complete sentence with brief context; 202 of the 233 questions have at least six usable sentence answers and enter the comparison. The label is the one of the entity answers, since the model’s knowledge does not change with the format.
- **Hearing statements.** Without the transcript, the model is told the topic of a PublicHearingBR hearing [6] and asked, in one short sentence, for the position of a named participant. There is no usable correctness label (Section 6.3).
- **Nulls.** From the Qwen hearing statements: a *half-contaminated* set keeps four of a prompt’s own samples and adds four drawn at random from the pooled samples of all prompts, and a *shuffled* set draws all eight from that pool.
- **Evidence swap.** The hearing prompt, now with five transcript sentences: in arm A, the five sentences that support the gold opinion; in arm B, five sentences from another, randomly drawn hearing. The answers of arm B are therefore confabulated by construction: produced from evidence unrelated to the question.
- **PublicHearingBR verification.** Human-labelled opinions, each with the 14 transcript sentences most similar to it by embedding cosine [13] (Section 7).

Table 3. Answer sets used in this report.

Setting	Sets	Samples	Generator	Label
TriviaQA, entity answers	233	8–10	Qwen2.5-7B	alias match
TriviaQA, sentence answers	202	6–8	Qwen2.5-7B	same as entity answers
Hearing statements	259	8	Qwen2.5-7B	—
Hearing statements	70	7–8	Llama-3.1-8B	—
Half-contaminated and shuffled nulls	2×259	8	Qwen2.5-7B	—
Evidence swap, arms A and B	2×80	7–8	Qwen2.5-7B	by construction
PublicHearingBR verification	837 opinions	14 sentences	—	human verdict

Pipeline. For each answer set:

1. run the NLI model on every ordered pair of answers, keeping the entailment and contradiction probabilities;
2. threshold entailment at 0.5, close it under transitivity, group mutually entailing answers into classes with masses p , and keep the order between classes;

3. threshold contradiction at 0.5 in either order to obtain raw conflict between classes;
4. close conflict under inheritance (Definition 1) and set self-conflicting classes aside;
5. enumerate the worlds (Proposition 3) and compute the number of worlds, WE and the conflict density; SE and SE_{core} come from the same classes.

No threshold was tuned: both are the values fixed in an earlier phase of the project. The contradiction probabilities come from the same forward passes that semantic entropy needs for entailment, so worlds add no model calls.

Statistics. On TriviaQA each question is scored by its negated uncertainty, and the AUROC measures how well the score predicts that the answer is correct; ties count one half. Differences are point estimates. Their 95% intervals are the 2.5 and 97.5 percentiles of 5,000 bootstrap resamples of the unit named in the text: questions for TriviaQA (paired between the two scores), prompts for the evidence swap, and hearings for PublicHearingBR, whose opinions from the same hearing are not independent. Proportions carry Wilson intervals.

6 Results on answer sets

6.1 Conflict carries information the order does not

On TriviaQA, contradiction fires on 30.0% of the unordered pairs of samples. Of the class pairs in conflict, 97.9% are incomparable in the entailment order, so conflict is not a restatement of the order. The NLI model almost satisfies inheritance by itself (mean closure cost 0.81% of class pairs per answer set) and rarely produces self-conflicting classes (on average 1.67% of the classes of an answer set). In all, 55.4% of the answer sets contain at least one conflict and 53.6% have more than one world (2.63 worlds on average).

The conflict relation means what its name says. We check this without any neural model, by matching each individual sample against the gold aliases: pairs of samples in conflict differ in correctness (one matches, the other does not) in 21.0% of cases, against 4.0% for pairs not in conflict (difference +0.170, 95% CI [+0.113, +0.229], resampling questions).

Table 4 splits the questions by the kind of diversity among their samples. Semantic entropy treats the two lower rows as the same kind of evidence, differing only in amount. Accuracy falls across the three rows. The fall from one class to several classes in one world is significant (+0.367 [+0.122, +0.600]); the fall from one world to several (+0.220 [−0.018, +0.459]) rests on only 18 one-world sets, and its interval includes zero. Correct answers come from sets with 1.61 worlds on average, wrong answers from sets with 3.53.

Table 4. Accuracy of the model’s answer by kind of diversity among its samples (TriviaQA, entity answers). Means of SE and WE per stratum.

Stratum	<i>n</i>	Accuracy [95% CI]	SE	WE
One class	90	81.1% [71.8, 87.9]	0.000	0.000
Several classes, one world	18	44.4% [24.6, 66.3]	0.626	0.000
Several classes, several worlds	125	22.4% [16.0, 30.5]	1.300	1.103

6.2 Worlds do not outperform semantic entropy where it works

Table 5 compares the scores as detectors of wrong answers. Semantic entropy reaches 0.811, reproducing the value of the earlier phase from the same cached samples (prediction P1). Neither order-aware nor conflict-aware corrections improve on it. The registered primary comparison, WE against SE, gives −0.0191 [−0.0445, +0.0057] (P3, not confirmed), and WE does no better than the core entropy (−0.0043 [−0.0277, +0.0185]). WE does not outperform the scalar conflict density either (+0.0221 [−0.0187, +0.0646]; P4, not confirmed), and adding it to the density by rank sum, the other arm of P4, which we ran only on 30 September 2026, gives +0.0203 [−0.0027, +0.0451]. Counting classes (0.809) is as good as any refinement. The conflation rate suggests why: on TriviaQA only 12.6% of the multi-class answer sets form a single world (9.8% contain no contradiction at all), so the classes semantic entropy counts are nearly always disagreements, and a conflict-aware score has little to correct.

Table 5. Detecting wrong answers on TriviaQA ($n = 233$, accuracy 46.8%). Left: AUROC. Right: paired differences with 95% intervals. LGU is our reimplementation of Dong et al. [3] from its description.

Score	AUROC	Comparison	Δ AUROC [95% CI]
Semantic entropy SE	0.811	WE – SE (P3)	–0.0191 [–0.0445, +0.0057]
Number of classes	0.809	SE _{core} – SE	–0.0148 [–0.0358, +0.0030]
LGU	0.805	WE – SE _{core}	–0.0043 [–0.0277, +0.0185]
Core entropy SE _{core}	0.796	WE – conflict density (P4)	+0.0221 [–0.0187, +0.0646]
World entropy WE	0.792	worlds – conflict density	+0.0214 [–0.0191, +0.0625]
Number of worlds	0.791	WE–LGU	–0.0135 [–0.0329, +0.0040]
Conflict density	0.770		

6.3 The conflation rate varies across settings

Table 6 applies the same code, NLI model and thresholds to the hearing statements and to the two nulls. The number of classes, worlds, entropies and densities are means over answer sets, which is why they are not integers.

Table 6. Conflict structure by setting (scripts `domain_conflict` and `intervals`). Means over answer sets, except n and the conflation rate, which is the share of multi-class sets forming a single world (Wilson 95% interval).

Setting	n	Classes	Worlds	>1 world	SE	WE	Density	Conflation
Hearing statements, Qwen	259	4.14	1.23	18.5%	1.112	0.125	0.103	78.5% [72.6, 83.4]
Hearing statements, Llama	70	6.11	1.87	50.0%	1.644	0.393	0.230	49.3% [37.8, 60.8]
TriviaQA, entity answers	233	3.28	2.63	53.6%	0.746	0.592	0.473	12.6% [8.1, 19.0]
Half-contaminated null	259	6.65	2.97	94.2%	1.800	0.933	0.424	5.8% [3.5, 9.3]
Shuffled null	259	7.91	4.06	98.5%	2.063	1.288	0.423	1.5% [0.6, 3.9]

We read the table in three steps. First, when the model is asked about a participant’s position without the transcript, 78.5% of the answer sets with several classes form a single world: the diversity that semantic entropy measures there (mean 1.112) is mostly elaboration and complementary detail, and the world entropy is about nine times lower (0.125). On TriviaQA the share is 12.6%. The Llama replication lies in between (49.3%), with more varied samples (6.11 classes per set against 4.14). A few of the one-world sets do contain a contradiction involving a self-conflicting class (15 of 175 for Qwen, 3 of 34 for Llama, 4 of 18 on TriviaQA); counting only sets with no contradiction at all gives 71.7%, 44.9% and 9.8%, in the same order. Second, the nulls behave as contamination should: replacing a prompt’s samples by samples drawn from the pool raises the number of worlds monotonically (1.23, 2.97, 4.06). Where the conflict density saturates (0.424 and 0.423), the number of worlds still separates the two nulls, so it carries information that the density does not. We had predicted that unrelated statements would rarely contradict each other and that the shuffled null would stay near one world (P7); instead, the NLI model reads many statements from different hearings as contradictory, such as two statements that name different states as recipients. Third, the diagnostics stay small: in the hearing settings, nulls and evidence-swap arms, the mean closure cost per answer set is 2.2 to 6.0% of class pairs and the mean self-conflict rate 3.2 to 5.5% of classes (on TriviaQA, 0.81% and 1.67%).

In our research log this pattern was called the *conflation law*: semantic entropy works where the diversity it counts is disagreement. The table alone cannot establish it, because the hearing statements offer no working reference: without the transcript the model almost never knows the answer, correctness hardly varies, and in an earlier phase of the project every score we tried there, semantic entropy included, was at chance. With one labelled setting, the conflation rate is a hypothesis about the scope of semantic entropy, which we test by intervention.

6.4 Raising the conflation rate lowers the AUROC of semantic entropy

If semantic entropy works where the diversity it measures is disagreement, changing only the form of the answers should move both. We asked the same TriviaQA questions for a complete sentence with brief context instead of a bare entity, keeping the label of the entity answers. Four predictions were registered in the script before the analysis (Section 8).

With sentence answers the conflation rate more than doubles, from 12.6% to 29.0% (+0.164 [+0.077, +0.250]; 9.6% to 24.3% counting only sets with no contradiction at all), less than the 40% we had predicted, and the AUROC of semantic entropy falls from 0.816 to 0.721 (–0.0953 [–0.1746, –0.0157];

Table 7. The same 202 TriviaQA questions and labels, answered with an entity or with a sentence (scripts `conflation_intervention` and `intervals`). Rows above the line are means over answer sets, except the conflation rate. Differences are paired over questions.

	Entity	Sentence	Sentence – entity [95% CI]
Classes	3.57	4.24	
Worlds	2.84	2.63	
Semantic entropy SE	0.840	1.106	
World entropy WE	0.667	0.645	
Conflation rate	12.6%	29.0%	+0.164 [+0.077, +0.250]
AUROC of SE	0.816	0.721	–0.0953 [–0.1746, –0.0157]
AUROC of WE	0.794	0.751	–0.0435 [–0.1187, +0.0270]
AUROC of SE _{core}	0.801	0.765	

Table 7). This is the one comparison in this report in which a change predicted from the conflation rate is observed on labelled data, with questions and labels held fixed. The sentence answers have at most eight samples per question, against ten for entity answers; restricting the entity answers to their first eight samples leaves semantic entropy at 0.819 and the drop at -0.0985 [$-0.1756, -0.0192$], so the smaller sample does not explain it. The world entropy falls less and ends above semantic entropy (0.751 against 0.721), but neither the difference between the two scores ($+0.0302$ [$-0.0219, +0.0837$]) nor the difference between their drops (-0.0518 [$-0.1096, +0.0044$]) is significant. The core entropy also falls less and reaches 0.765, the highest of the three with sentence answers; we had not registered this comparison and report it as an observation ($WE - SE_{core} = -0.0136$ [$-0.0529, +0.0269$]). The sentence answers still carry the entity: a gold alias appears in 47.7% of them, and the per-question share of such answers correlates with the label at $+0.692$.

7 Negative results

Contradictions in the evidence do not indicate hallucination. The main paper’s task is to decide whether an opinion attributed to a participant is supported by the transcript. The consistency structure can be carried to that task only through the evidence. For an opinion, take the 14 transcript sentences most similar to it and keep those that entail it with probability at least 0.35 (its *support*); order them by entailment and relate them by contradiction as in Section 3, with equal masses. We predicted (P8) that an opinion whose support spans two or more worlds, that is, support that cannot all be true, would more often be unsupported. We tested this on the 837 opinions from 40 hearings for which these features were computed (93 of them unsupported). The prediction fails (Table 8): the number of worlds in the support, its world entropy and the flag for support spanning several worlds all score below chance, with intervals that resample hearings and exclude 0.5. The reason is the size of the support. An opinion with no supporting sentence (21.6% of them) has no world at all, the number of worlds grows with the number of supporting sentences (correlation $+0.69$), and opinions with more support are less often unsupported (support size alone scores 0.602). Among the 495 opinions with at least two supporting sentences (44 unsupported), comparing only opinions with the same number of supporting sentences, the number of worlds scores 0.512 [$0.456, 0.571$]; its entropy gives the same value and the flag 0.510 [$0.455, 0.568$]. We detect no information in the coherence of the evidence beyond its amount. A plausible reading is that well-discussed opinions attract more evidence and, with it, more contradictions. Contradiction between the opinion and any retrieved sentence is at chance.

None of the self-consistency quantities we tested detects answers built on misleading evidence. In the evidence swap, the answers of arm B share 20 times less vocabulary with the gold opinion than those of arm A (Jaccard overlap of content words 0.008 against 0.167; paired difference $+0.159$ [$+0.132, +0.185$]), so they are confabulated. As registered (P6), no quantity separates the arms (paired differences A–B over the 80 prompts): number of worlds $+0.138$ [$-0.037, +0.325$], world entropy $+0.080$ [$-0.015, +0.172$], conflict density -0.009 [$-0.089, +0.068$], semantic entropy $+0.012$ [$-0.160, +0.184$]. As far as these measures can tell, a model given the wrong evidence answers as consistently as one given the right evidence; we found no quantity, with or without worlds, that separates the two.

Table 8. Consistency features against the human verdict in PublicHearingBR ($n = 837$ opinions from 40 hearings, 93 unsupported; scripts `incoherent_support` and `intervals`). AUROC for detecting unsupported opinions, with 95% intervals that resample hearings.

Feature	AUROC	[95% CI]
Maximum entailment of the opinion (baseline)	0.598	[0.527, 0.676]
Size of the support (threshold 0.35)	0.602	[0.542, 0.665]
Maximum embedding cosine	0.544	[0.451, 0.645]
Contradiction between opinion and evidence	0.504	[0.428, 0.580]
World entropy of the support	0.459	[0.424, 0.497]
Support spans several worlds (flag)	0.457	[0.424, 0.494]
Number of worlds in the support	0.433	[0.374, 0.489]

8 Predictions and outcomes

Predictions P1–P8 were registered on 3 August 2026, before any AUROC of the new quantities was computed; P3 and P4 were the confirmatory comparisons, and the rest were secondary or exploratory. The predictions for the sentence-answer intervention were registered in the header of `conflation_intervention` before its analysis. Table 9 lists every outcome.

Table 9. Registered predictions and their outcomes. *Confirmatory.

	Prediction	Outcome
P1	SE reproduces 0.811 from the cached samples	Confirmed (0.811)
P2	Most conflicts join incomparable classes; closure cost below 10%; self-conflict below 5%	Confirmed (97.9%; 0.81%; 1.67%)
P3*	WE loses less to SE than SE_{core} does (-0.0149); hoped for $WE \geq SE$	Not confirmed (-0.0191 [-0.0445 , $+0.0057$])
P4*	Worlds or WE outperform the conflict density, or add to it in combination	Not confirmed ($+0.0221$ [-0.0187 , $+0.0646$]; combination $+0.0203$ [-0.0027 , $+0.0451$], run late)
P5	Among the 28 answer sets with redundant classes (core smaller than the class count), wrong answers have more worlds (exploratory)	Direction as predicted, not significant (AUROC of worlds 0.655 [0.417, 0.873])
P6	No world quantity separates the evidence-swap arms	Confirmed (Section 7)
P7	Hearing statements mostly form one world; so does the shuffled null	First half confirmed (81.5% of sets); second refuted (4.06 worlds)
P8	Support spanning several worlds concentrates hallucinations	Refuted: below chance (0.43–0.46) through support size; 0.512 [0.456, 0.571] at equal size
h5 (i)	Conflation rises to at least 40% with sentence answers	Not confirmed: it rises ($+0.164$ [$+0.077$, $+0.250$]), but to 29.0% [22.7, 36.2]
h5 (ii)	The AUROC of SE falls with sentence answers	Confirmed (-0.0953 [-0.1746 , -0.0157])
h5 (iii)	$WE \geq SE$ with sentence answers	Not significant ($+0.0302$ [-0.0219 , $+0.0837$])
h5 (iv)	SE falls more than WE	Direction as predicted, not significant (-0.0518 [-0.1096 , $+0.0044$])

9 Limitations

One verifier. Every relation in this report comes from one multilingual NLI checkpoint at fixed thresholds. The conflict relation is only as good as that model; the nulls show that it reads many unrelated hearing statements as contradictory, and we report closure cost and self-conflict rather than hide them.

Few generators and settings. Two small open models (Qwen2.5-7B and Llama-3.1-8B), one question-answering benchmark and one task on hearings. The conflation rate is a property of a combination of model, prompt and domain, and the Llama replication already moves it from 78.5% to 49.3%. More generators and domains are needed before the pattern can be called general.

Labels and causality. Only TriviaQA has correctness labels, and they come from alias matching. The evidence that the conflation rate governs the AUROC of semantic entropy rests on a single intervention with 202 questions, which changes the format of the answers and, with it, possibly other properties of the samples; within it, the advantage of world entropy is not significant. No comparison was corrected for multiplicity: P3 and P4 were designated as confirmatory in advance, and every other comparison should be read as exploratory.

Scope of the verification test. The PublicHearingBR analysis covers the 837 opinions of an earlier, interrupted collection, not the 3,630 opinions of the main paper, and uses the 14 most similar sentences rather than the attributed speaker’s turns.

Comparators. The core entropy is our own construction, and LGU is a reimplementation from its description; neither is an official implementation.

A corrected script. While preparing this report we found that the script for Table 6 treated two identical samples as different classes. The corrected values are reported. The correction affects the settings with repeated samples (the Qwen hearing statements, the half-contaminated null and the evidence-swap arms): it lowers their mean number of classes and SE (for Qwen, from the previously reported 4.31 and 1.151 to 4.14 and 1.112), changes their conflict densities, verifier diagnostics and evidence-swap differences slightly, leaves the numbers of worlds and WE unchanged, and moves the conflation rates by at most 0.4 percentage points.

10 Conclusion

Semantic entropy counts how many different meanings a model produces, but not whether those meanings exclude one another. The contradiction probabilities that the NLI model already computes are enough to tell the two apart: once conflict is inherited along entailment, the maximal consistent subsets of an answer set are its maximal independent sets, which a standard clique algorithm enumerates exactly. On TriviaQA, where almost all diversity among answers is disagreement, the distinction adds nothing to semantic entropy as a detector, but it helps explain why semantic entropy works there. When Qwen states the positions of hearing participants without the transcript, most multi-class answer sets form a single world (about half with Llama), and asking TriviaQA questions for full sentences both raises the share of such sets and lowers the AUROC of semantic entropy. Because the conflation rate needs no labels, it can be measured in a new setting before an uncertainty method is trusted there. Our results do not show that world entropy should replace semantic entropy, and they do not carry over to the verification task of the main paper, where we detect no information in the contradictions within the evidence beyond its amount.

Reproducibility. The directory `code/` next to this report’s directory contains every script and cached intermediate result, and `code/run_all.sh` runs them. The scripts `event_structure`, `triviaqa_worlds`, `domain_conflict`, `conflation_intervention` and `intervals` run offline in about a minute on a laptop and print every number in this report; `incoherent_support` also needs the PublicHearingBR data, as in the main repository. No language model or NLI call is needed unless the caches are deleted. `README.md` lists the commands and the expected output.

References

- [1] Coen Bron and Joep Kerbosch. Algorithm 457: Finding all cliques of an undirected graph. *Communications of the ACM*, 16(9):575–577, 1973.
- [2] Longchao Da, Tiejun Chen, Lu Cheng, and Hua Wei. LLM uncertainty quantification through directional entailment graph and claim level response augmentation, 2024. arXiv:2407.00994.
- [3] Yanni Dong, Minghua Liu, Meiling Zhu, Xiaowei Huang, and Lijun Zhang. Beyond semantic equivalence: Logical graphs for LLM uncertainty quantification, 2026. arXiv:2607.16868.
- [4] Clifford H. Dowker. Homology groups of relations. *Annals of Mathematics*, 56(1):84–95, 1952.
- [5] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.
- [6] Leandro Carísio Fernandes, Guilherme Zeferino Rodrigues Dobins, Roberto Lotufo, and Jayr Alencar Pereira. PublicHearingBR: A brazilian portuguese dataset of public hearing transcripts for summarization of long documents, 2024. arXiv:2410.07495.
- [7] Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Proc. ACL*, 2017.
- [8] Dmitry Kozlov. *Combinatorial Algebraic Topology*, volume 21 of *Algorithms and Computation in Mathematics*. Springer, 2008.
- [9] Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *ICLR*, 2023.

- [10] Moritz Laurer, Wouter van Atteveldt, Andreu Casas, and Kasper Welbers. Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and BERT-NLI. *Political Analysis*, 32(1): 84–100, 2024.
- [11] Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *EMNLP*, 2023.
- [12] Alexander Nikitin, Jannik Kossen, Yarin Gal, and Pekka Marttinen. Kernel language entropy: Fine-grained uncertainty quantification for LLMs from semantic similarities. In *NeurIPS* 37, 2024. arXiv:2405.20003.
- [13] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *EMNLP*, 2019.
- [14] Robert E. Stong. Finite topological spaces. *Transactions of the American Mathematical Society*, 123(2):325–340, 1966.
- [15] Glynn Winskel. An introduction to event structures. In *Linear Time, Branching Time and Partial Order in Logics and Models for Concurrency*, volume 354 of *LNCS*, pages 364–397. Springer, 1988.